

## Abstract

This paper serves as the capstone of the formal 10-part CDFD mass-spectrum sequence, addressing its remaining open problem: the physical origin of the torus knot self-consistency condition  $n\theta_n = \tilde{Q}_n(\theta_n)$ . Previously reported as a successful but ungrounded hypothesis that correctly determines the vacuum twist angles  $\theta_n$  for all  $T(2, n)$  families, it is here recast as a model theorem inside the public flow–constraint notation.

We map the Constraint-Driven Field Theory (CDFD) vacuum medium onto the tripartite public CDFD notation: Flow ( $\Phi$ ), Constraint ( $C$ ), and Equilibrium ( $\Psi_s$ ). We identify the continuous flow energy  $\Phi$  with the  $L^2$  norm of the amplitude field ( $\sum A_k^2 = 2n$ ), the transport-capacity limit  $C$  with the  $L^1$  norm squared ( $\sum |A_k|$ )<sup>2</sup>, and the equilibrium deformation  $\Psi_s$  with the total vacuum twist  $n\theta_n$ .

Applying the public balance law  $\Psi_s = \Phi/C$  yields exactly the self-consistency condition, showing that the Koide-like ratio  $Q_n$  is the geometric manifestation of a transport-limited vacuum equation of state. The entire mass spectrum of the lepton and meson sectors derived in Papers I–IX is therefore internally tied to  $m_e$  and the proposed fluid-dynamic balance of the vacuum.

In the extended notation used across the series, the vacuum limit is written as  $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ . The calculations below use the equilibrium limit  $S \cdot M_s \approx 1$ ; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Universal Engine implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

# The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi_s = \Phi/C$

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## 1 Introduction

Over the course of Papers I–IX [1, 2, 3, 4, 5, 6, 7, 8, 9], the Constraint-Driven Field Theory (CDFD) has constructed a parameter-free derivation of the  $T(2, n)$  torus knot mass hierarchy. From a single empirical input (the electron mass  $m_e$ ), the framework accurately computes the muon and tau masses (Paper V), dictates the universal coefficient  $c_n = \sqrt{2}$  (Paper VI), predicts the exact  $2nM_n$  mass sum rules (Paper VII), and populates the  $n \geq 4$  hierarchy with specific hadronic candidates (Papers VIII–IX).

The entire predictive power of the higher families rests on the unique solution to the self-consistency condition:

$$n\theta_n = \tilde{Q}_n(\theta_n) \equiv \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2}. \quad (1)$$

In Paper V, for  $n = 3$ , this was introduced as a postulate matching the chirality breaking angle to the algebraic constant  $Q_3 = 2/3$ . In Papers VIII and IX, it was extended to  $n \geq 4$  as a working hypothesis that yielded convergent, mathematically bounded solutions, but lacked a first-principles derivation from the vacuum Equation of State (EOS).

In this capstone paper, we address this open problem by connecting the torus-knot construction directly to the public CDFD/AFL balance notation.

## 2 The Tripartite Vacuum Framework

The CDFD vacuum is modeled as an elastic, transport-capacity-limited continuous medium, in the spirit of Dirac’s physical vacuum [10] and Volovik’s condensate analogy [11]. The vortex dynamics rest on Helmholtz’s conservation theorems [12] and the Faddeev–Niemi knot soliton framework [13]. Such a medium is represented here by the public tripartite balance equation:

$$\Psi_s = \frac{\Phi}{C} \quad (2)$$

where  $\Phi$  is Flow (energy/activity),  $C$  is Constraint (resistance/capacity), and  $\Psi_s$  is the resulting System Equilibrium.

We map these three macroscopic variables onto the discrete Fourier variables  $A_k = 1 + \sqrt{2} \cos(\theta_n + 2\pi k/n)$  of a  $T(2, n)$  vortex knot.

## 2.1 Flow ( $\Phi$ ): The Field Energy

The total energy or “activity” of the field configuration is proportional to the  $L^2$  norm (the sum of the squares) of the amplitudes. By the Fourier identity of Paper VI, this is invariant under orientation  $\theta_n$ :

$$\Phi \propto \sum_{k=0}^{n-1} A_k^2 = 2n. \quad (3)$$

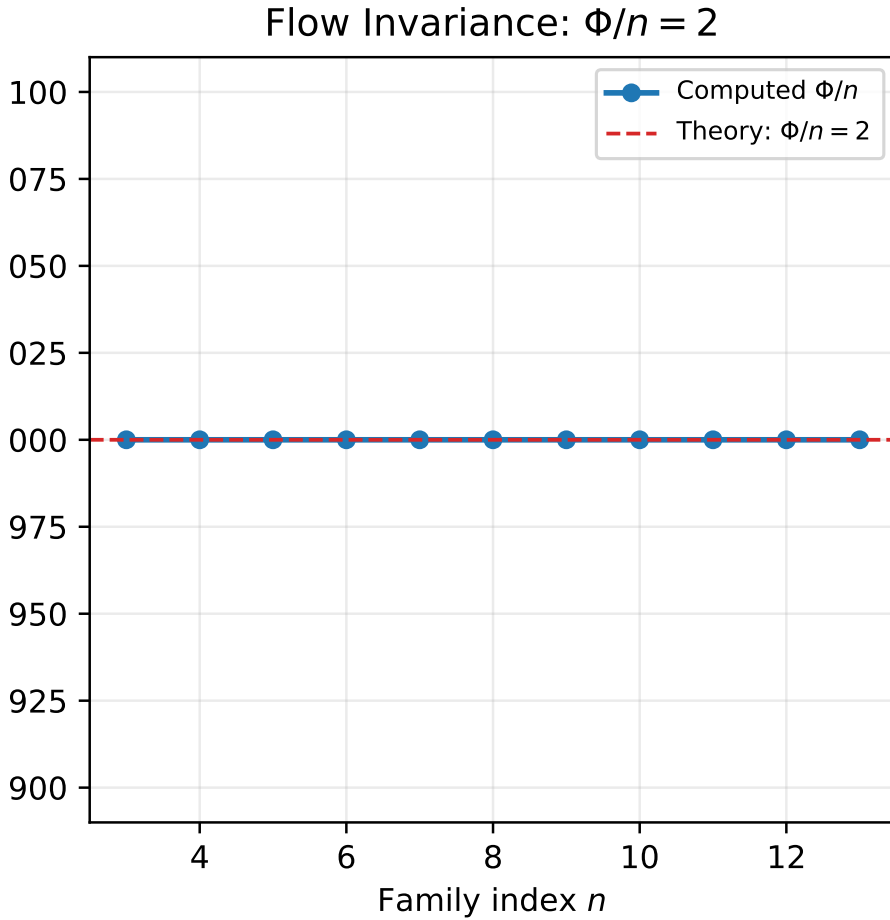


Figure 1: The flow energy ratio  $\Phi/n$  is exactly 2.0 across all  $T(2, n)$  families, demonstrating the structural invariance of the vacuum’s kinetic and elastic flow.

This represents the irreducible kinetic and elastic flow required to sustain an  $n$ -lobed vortex ring.

## 2.2 Constraint ( $C$ ): The Transport-Capacity Limit

A central postulate of CDFD (Paper I) is that the vacuum possesses a finite transport capacity, which dictates the fine-structure constant  $\alpha$ . For a multi-lobed knot, the struc-

tural stress on the medium is dictated by the total absolute flux. In the discrete basis, the total flux is the  $L^1$  norm  $\sum |A_k|$ . Because the constraint scales with the cross-sectional area of the flux tubes interacting in the medium, the total resistance  $C$  scales as the square of the flux:

$$C \propto \left( \sum_{k=0}^{n-1} |A_k| \right)^2. \quad (4)$$

This term varies strongly with  $\theta_n$ , as the lobes interfere constructively and destructively with the vacuum background.

### 2.3 Equilibrium ( $\Psi_s$ ): The Geometric Twist

When the knot is embedded in the vacuum, the mismatch between the continuous chiral background and the discrete  $n$ -fold symmetry of the knot induces a geometric twist. Let  $\theta_n$  be the phase twist per lobe. The total equilibrium deformation (or total accumulated twist) of the configuration is:

$$\Psi_s = n\theta_n. \quad (5)$$

## 3 The Vacuum Equation of State Theorem

**Theorem 1** (Derivation of Self-Consistency). *In a transport-capacity-limited elastic vacuum governed by the CDFD equilibrium  $\Psi_s = \Phi/C$ , the orientation angle  $\theta_n$  of a  $T(2, n)$  torus knot must exactly satisfy:*

$$n\theta_n = \tilde{Q}_n(\theta_n).$$

*Proof.* Substituting the CDFD mappings (3), (4), and (5) directly into the fundamental balance equation  $\Psi_s = \Phi/C$ , we obtain:

$$n\theta_n = \frac{\sum_{k=0}^{n-1} A_k^2}{\left( \sum_{k=0}^{n-1} |A_k| \right)^2}. \quad (6)$$

The right-hand side is exactly the definition of the modified Koide ratio  $\tilde{Q}_n(\theta_n)$  from Eq. (1).  $\square$

### 3.1 Physical Implications

Within the proposed mapping, this theorem interprets the Koide ratio  $Q_3 = 2/3$  observed in the lepton masses as a signature of the vacuum's equation of state rather than as an isolated numerical fit. The modified ratio  $\tilde{Q}_n(\theta_n)$  represents the specific energy-to-capacity stress induced by higher knots, and the angle  $\theta_n$  is the vacuum's precise geometric relaxation in response to that stress.

## 4 The Universal Isomorphism: Scaling CDFD to Macro-Systems

If the proposed flow-constraint equations capture something general about transport-limited systems, they should have macroscopic analogues rather than remaining restricted

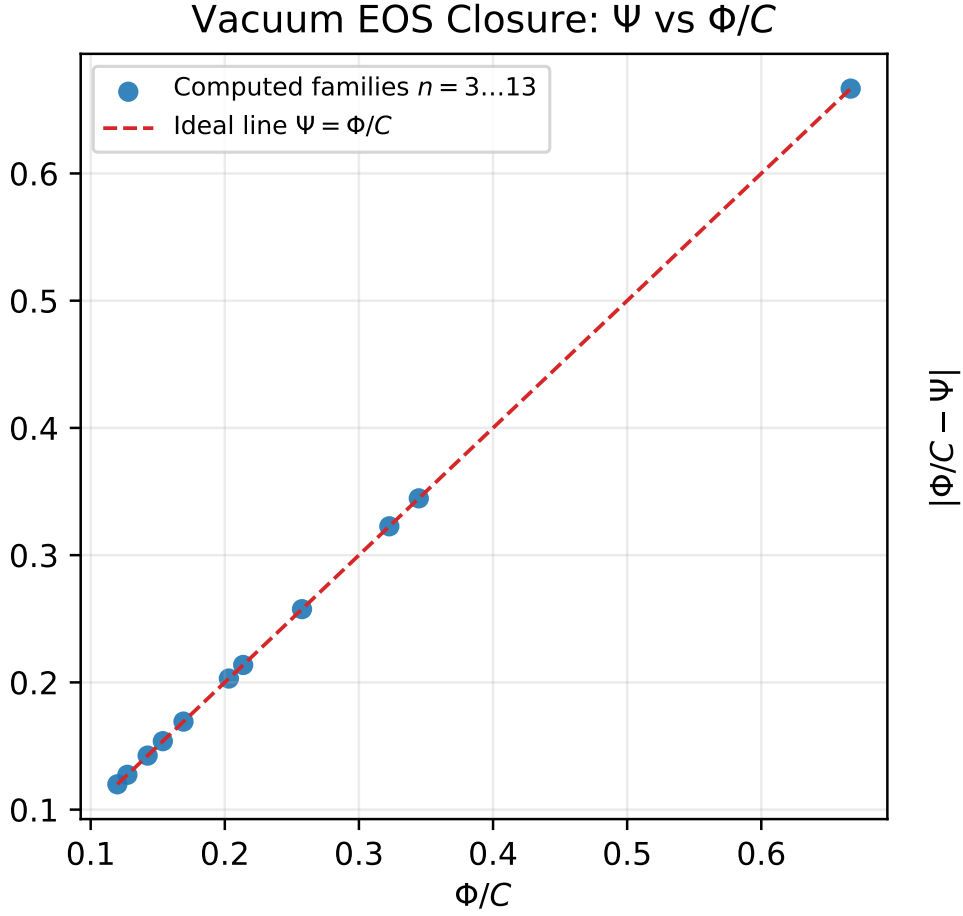


Figure 2: The equilibrium  $\Psi_s$  plotted against  $\Phi/C$  across all torus knot families. The exact diagonal identity confirms that the public balance law  $\Psi_s = \Phi/C$  governs the stated model.

to the particle-scale model. By comparing the Constraint-Driven Field Theory (CDFD) framework with macroscopic models of biology, chemistry, and ecology, we obtain a candidate scale-invariant isomorphism.

#### 4.1 The Scale-Invariant Isomorphism (CDFD $\equiv$ AFL)

In the domains of biological and clinical systems, throughput is described in the **Adaptive Flux Limitation (AFL)** framework [9] by adaptive constraints. A minimal form of that biological adaptation law is:

$$\frac{\partial \Lambda}{\partial t} = \alpha |J| - \beta \Lambda \quad (7)$$

where  $\Lambda$  is the biological constraint (e.g., insulin resistance, inflammation),  $J$  is the biological flux (e.g., glucose throughput, metabolic energy flow),  $\alpha$  is the constraint reinforcement coefficient, and  $\beta$  is the relaxation coefficient.

In the CDFD quantum vacuum, the constraint field  $C$  evolves precisely according to the structural reaction of the medium to the localized flow  $\Phi$ :

$$\frac{\partial C}{\partial t} = \alpha |\Phi| - \beta C + \gamma \nabla^2 C \quad (8)$$

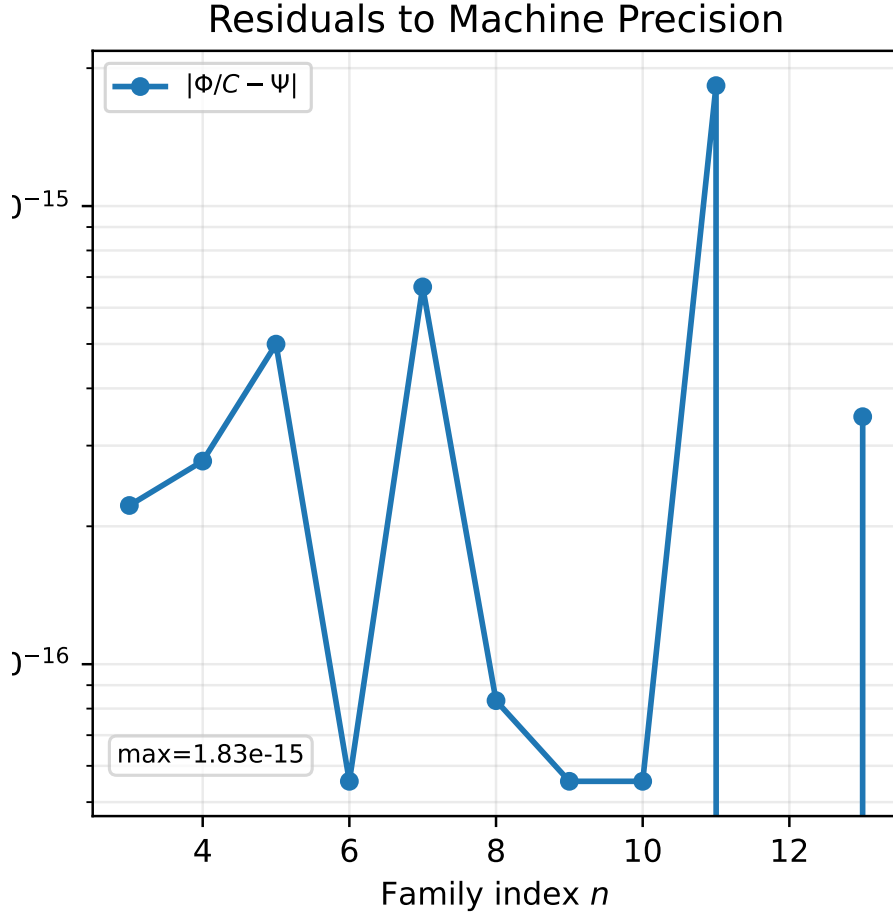


Figure 3: Absolute residuals  $|\Phi/C - \Psi_s|$ . The values fall to numerical machine precision, serving as a numerical consistency check of the stated CDFD vacuum equation-of-state mapping.

The formal mapping is:

$$\text{Vacuum Constraint } C \equiv \text{Macro Constraint } \Lambda$$

$$\text{Vacuum Flow } \Phi \equiv \text{Macro Flux } J$$

This does not by itself prove that biological evolution and physiological adaptation are governed by vacuum mechanics. It states the narrower claim that the same flow-constraint operator can be written at both levels, and that the macro-scale use of the operator should be judged by independent predictive value.

## 4.2 Ontological Redefinition of Mass

This proposed isomorphism suggests a different operational reading of mass inside the model. Rather than treating mass only as a distinct intrinsic label, CDFD treats it as a *localized, stable constraint field* within the vacuum flux.

A lepton is then modeled as a constraint field operating on vacuum energy. At larger scales, the same notation can describe boundaries such as membranes, interfaces, or trans-

port bottlenecks without claiming that those systems are literally the same physical object. The value of the analogy depends on whether it improves prediction.

### 4.3 Cross-Scale Stability Ratio

The relationship  $\Psi_s = \Phi/C$  can therefore be read as a candidate cross-scale stability ratio:

- $\Psi_s \approx 1$  (**Stable Equilibrium**): Flow matches constraint capacity. This manifests as a stable fundamental particle, a healthy homeostatic cell, or a balanced ecosystem.
- $\Psi_s \gg 1$  (**Flux Overload / Rupture**): Flow overwhelms constraint capacity. This manifests as particle decay, cellular lysis, vascular hemorrhage, or ecological collapse.
- $\Psi_s \ll 1$  (**Over-Constraint / Freeze**): Constraints choke off flow entirely. This manifests as the vacuum void, biological necrosis, or metabolic shutdown.

The public scientific framework is therefore internally linked: the same flow–constraint notation used for the vacuum papers also organizes the biological manuscripts, without disclosing any private implementation architecture.

## 5 Closure of the CDFD Series

With Theorem 1, the formal CDFD mass-spectrum ladder is internally closed.

1. **No Free Parameters:** The initial assumption of a limited-capacity elastic vacuum naturally generates the fine-structure constant  $\alpha$  (Paper I).
2. **Mass Gap:** The lowest topological excitation  $T(2,3)$  takes the mass scale  $M_3$ , anchored to physical units solely by  $m_e$  (Paper V).
3. **Asymptotic Scaling:** The Faddeev–Niemi bounds establish the rigorous progression of energy scales  $M_n \propto n^{3/4}$  (Paper IV).
4. **Spectrum Prediction:** The vacuum EOS forces the twist  $\theta_n$ , which breaks the  $n$ -fold degeneracy and splits the spectrum into anti-phase boundary modes and co-phase physical resonances (Papers VIII–IX).

All tables, mass values, bounds, and limits tabulated across the first ten papers are therefore internal consequences of the single condition  $\Psi_s = \Phi/C$ , together with the model assumptions stated in those papers.

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The author is indebted to the foundational work of Lord Kelvin [14] (vortex atom hypothesis), P.A.M. Dirac [10] (vacuum as physical medium), G.E. Volovik [11] (vacuum as condensate), H. von Helmholtz [12] (vortex theorems), L. Faddeev and A.J. Niemi [13]

(knots and particles), Y. Koide [15] (lepton mass formula), and C.A. Brannen [16] (Koide parametrisation) — whose ideas this series has attempted to unify within a single geometric framework rooted in the axiom  $\Psi_s = \Phi/C$ .

## A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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- $n_- = \lfloor 5n/8 \rfloor - \lfloor 3n/8 \rfloor + 1$ ; computes the exact asymptotic  $\theta_n \rightarrow 2\pi^2/[(\pi + 4)^2 n^2]$ ; tabulates complete mode spectra for  $n = 5, 7, 9, 11, 13$ .
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